

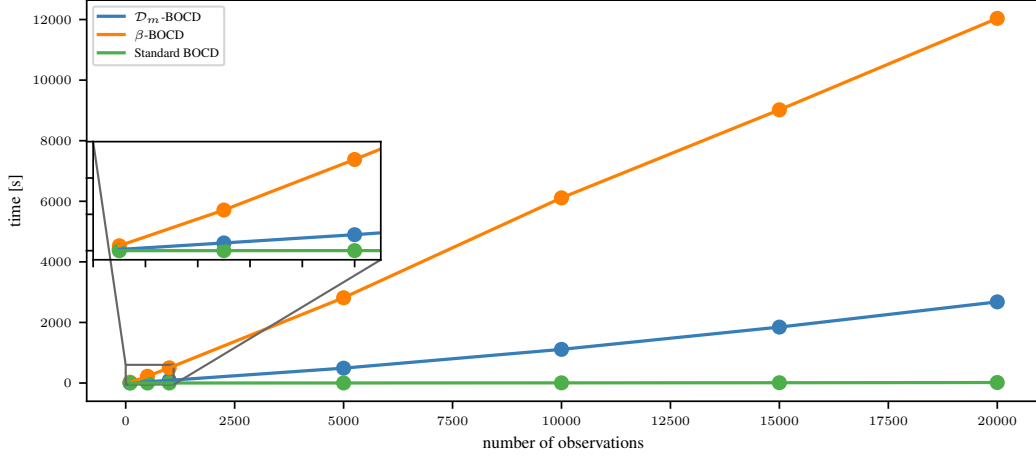


Figure 9. Overall time in seconds versus the number of observations. **blue** line for  $\mathcal{D}_m$ -BOCD, **orange** line for  $\beta$ -BOCD, and **green** line for standard BOCD. We observe that our method is faster than  $\beta$ -BOCD but is slower than the standard BOCD.

### C.2. Varying $m$ , $k$ , and outliers intensity.

We have demonstrated that our method is insensitive to the scale of the outliers. To do so, we compare the performance of the standard BOCD with the  $\mathcal{D}_m$ -BOCD in different contaminated datasets: while everything else stays the same, we scale the contamination points. We generate 600 samples with 6 CPs at  $T \in \{200, 400\}$ , and 3 contaminated point at  $T \in \{100, 300, 500\}$ . We contaminate the data by adding (or subtracting) 0, 5, 10, 20, and 40. For the  $\mathcal{D}_m$ -BOCD, we choose two different  $m$  functions:  $m$  as proposed to assure robustness, and  $m = I_d$ . In Figure 10, we observe that both Standard BOCD and  $\mathcal{D}_m$ -BOCD with  $m = I_d$  mistakenly label outliers as CPs, while  $\mathcal{D}_m$ -BOCD with  $m$  robust is robust and identifies lasting changes.

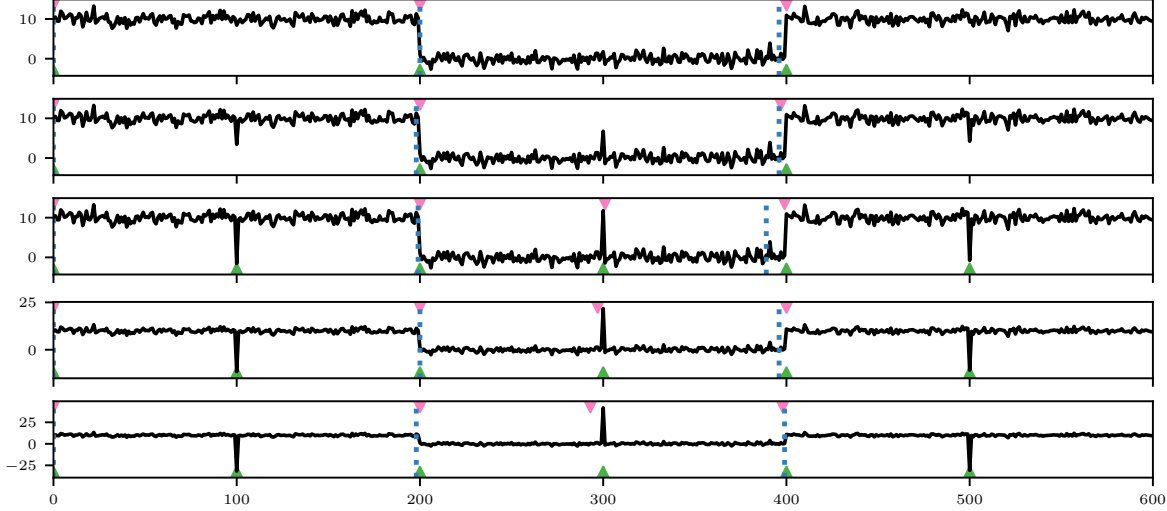


Figure 10. Contamination point scaled by 0, 5, 10, 20, and 40, from top to bottom. MAP segmentation indicated by **blue** dashed lines for  $\mathcal{D}_m$ -BOCD with  $m$  robust, **▼** for  $\mathcal{D}_m$ -BOCD with  $m = I_d$ , and **▲** for standard BOCD. Both Standard BOCD and  $\mathcal{D}_m$ -BOCD with  $m = I_d$  mistakenly label outliers as CPs, while  $\mathcal{D}_m$ -BOCD with  $m$  robust is robust and identifies lasting changes.

Finally, for contamination scaled by 10, we compared the performance and wall-clock time for  $k \in \{1, 50, 600\}$ . In Figure 11, we observe that for  $k = 1$  the method cannot detect any changepoint, while for  $k = 50$  and  $k = 600$ , the method successfully identifies the changepoints. However, in Table 2 for  $k = 600$ , the method takes 6x more time. As we discussed in the main paper,  $k$  is a parameter in all variants of BOCD and will make all algorithms more expensive if increased.

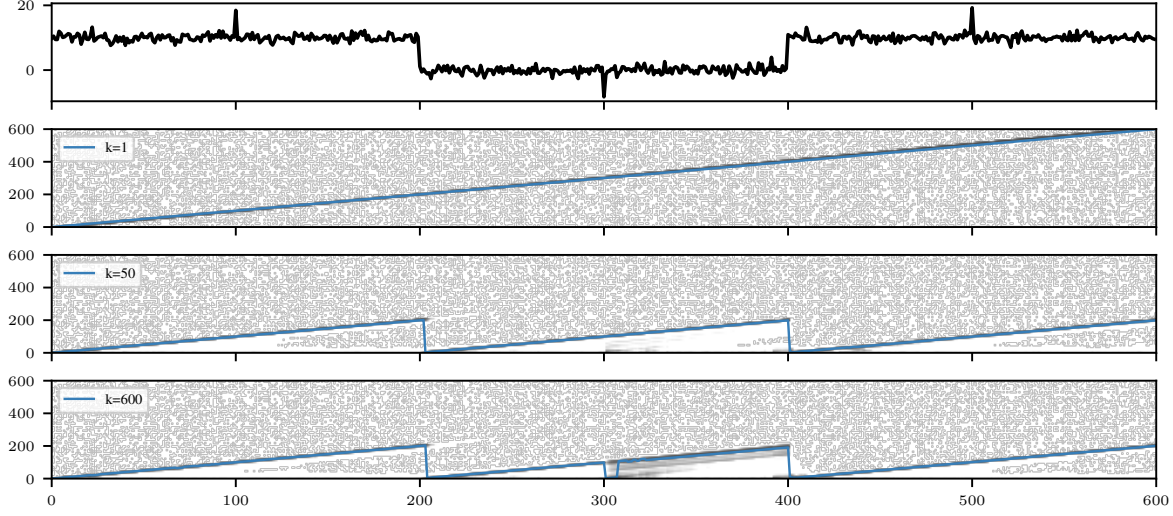


Figure 11. MAP segmentation indicated by blue dashed lines for  $\mathcal{D}_m$ -BOCD with  $m$  robust. For  $k = 1$  the method cannot detect any changepoint, while for  $k = 50$  and  $k = 600$ , the method successfully identifies the changepoints

| k   | time [s] |
|-----|----------|
| 1   | 1        |
| 50  | 38       |
| 600 | 236      |

Table 2. Wall-clock time for varying  $k$

### C.3. Accuracy and detection delay

We quantify the method’s performance advantage by comparing detection delay and accuracy on artificially generated data with outliers. In particular, We generate 600 samples with 2% of outliers and 6 CPs; then, we report the positive predictive value (PPV), true positive rate (TPR), and the detection delays. These metrics are given by

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}},$$

where TP, FP, and FN are true positives, false positives, and false negatives, respectively. We say the method detects a TP changepoint when the changepoint detected for the method is in a small neighbourhood of the true CP. In order to measure how far the predicted CP is from the true CP, we measure the time difference between both and call it detection delay. For both PPV and TPR, the nearest to 1, the better. For delays, the lower, the better. We run the experiment 10 times, varying the position of the outliers. The following table shows the mean and standard deviation for each metric:

| Method        | PPV                | TPR               | Delays            |
|---------------|--------------------|-------------------|-------------------|
| DSM-BOCD      | <b>0.907±0.154</b> | <b>0.883±0.13</b> | 1.643±0.475       |
| Standard BOCD | 0.6±0.128          | 0.833±0.149       | <b>1.05±1.545</b> |

Table 3. Performance indices-mean and standard deviation-using the positive predictive value (PPV), true positive rate (TPR), and the detection delays over 10 realisations. For PPV and TPR, the nearest to 1, the better. For delays, the lower, the better.

In Table 3, we observe  $\mathcal{D}_m$ -BOCD has a significantly better performance with respect to PPV, meaning that the rate of false positives is lower than standard BOCD. This is a consequence of the robustness of our method. Moreover, we see a similar result concerning TPR, which measures the amount of changepoint not detected for the methods. Overall, this means that our method detects the same amount of TP as the standard BOCD while not detecting many FP, showing the strength of our method. Lastly, the detection delay shows that in spite of being robust to outliers,  $\mathcal{D}_m$ -BOCD does not cause any delay in the detection of CP.

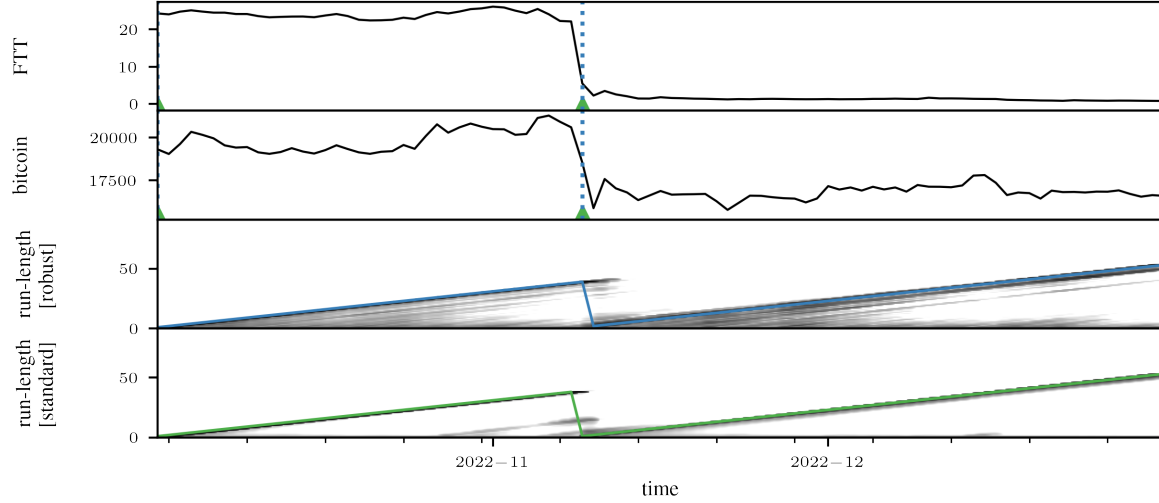


Figure 12. Maximum a posteriori (MAP) segmentation. In **blue**  $\mathcal{D}$ -BOCD segmentation using dashed lines. In **green** Standard BOCD segmentation using  $\blacktriangle$  marks. In addition we plot the run-length posteriors of robust  $\mathcal{D}_m$ -BOCD algorithm with most likely run-length in **blue** and of standard BOCD in **green**. We observe that robustness does not lead to increased CP detection latency: both methods detect the CP at the same time.

#### C.4. Twitter flash crash & Cryptocrash

In the Twitter flash crash experiment, we model the data with a Gaussian distribution, modelling its natural parameters. For  $\mathcal{D}_m$ -BOCD, we use a conjugate squared exponential prior  $r$  with parameters  $\mu = (0, 1)^\top$ , and  $\Sigma$  a diagonal matrix so that  $\text{diag}(\Sigma) = (10, 1)$  for the natural parameters of said Gaussian. For standard BOCD, we use a Normal-inverse gamma prior with parameters  $\mu_0 = 0$ ,  $\nu = 1$ ,  $\alpha = 2$  and  $\beta = 10$ . We use the first 50 observations to select  $\omega$  as in Section 3.4, with an obtained value of  $\omega^* \approx 0.0001$ .

In the Cryptocrash experiment, we model the data with a multivariate Gaussian distribution modelling its natural parameters, and use conjugate squared exponential prior with parameters  $\mu = (0, 1, 0, 1)^\top$ , and  $\Sigma$  diagonal matrix such that  $\text{diag}(\Sigma) = (2, 1, 2, 1)$  for the  $\mathcal{D}_m$ -BOCD. For standard BOCD, we model mean and variance instead, and use a normal-inverse-Wishart prior with parameters  $\nu = 0$ ,  $\kappa = 1$ ,  $\mu = (0, 0)$  and  $\Psi$  diagonal matrix such that  $\text{diag}(\Psi) = (1, 1)$ . we manually fix  $\omega = 0.01$ . Figure 12 shows the run-length posteriors and the Maximum A Posteriori (MAP) segmentations produced by each method. Since the most likely run-lengths are virtually identical, the below plot also shows that in spite of being robust to outliers,  $\mathcal{D}_m$ -BOCD does not cause any delay in the detection of CPs.

#### C.5. Well-log

We model the data with a Gaussian distribution, modelling its natural parameters and using a conjugate squared exponential prior with parameters  $\mu = (0, 10)^\top$ , and  $\Sigma$  diagonal matrix such that  $\text{diag}(\Sigma) = (100, 100)$ . Figure 13 shows the run-length and the segmentation of each method. We use the first 200 observations to select  $\omega$  as in Section 3.4, with an obtained value of  $\omega^* \approx 0.0004$ .

#### C.6. Multivariate synthetic data

We generate 1000 samples from a time series with CPs at  $t = 250, 750$ . Conditional on the CPs, the data is generated independently from an exponential in the first and Gaussian distribution in the second dimension. Since both dimensions are exponential family members, their joint distribution is too. On the natural parameters of this joint distribution, we place a conjugate squared exponential prior with parameters  $\mu = (1, 0, 0.5)^\top$ , and  $\Sigma$  diagonal matrix such that  $\text{diag}(\Sigma) = (1, 1, 0.2)$ . As we do not compare against BOCD on this data set, we simply fix  $\omega^* = 0.15$ .